INCH-POUND

A-A-59247A
14 September 2004
SUPERSEDING
A-A-59247
16 June 1998

COMMERCIAL ITEM DESCRIPTION

DRIVE, POWER, PORTABLE; FOR PIPE AND BOLT THREADING, ELECTRIC (WITH ACCESSORIES)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers hand-held and stand-mounted power drives powered by integral or separately mounted electric motors for cutting threads on pipe and bolt stock. Included are optional items to tailor the power drive to perform various functions. The power drives provide rotational power to turn pipe and bolt stock or turn geared threaders for cutting, threading, or reaming pipe. Type I is a hand-held drive that can also be adapted for turning large valves and geared threaders. Type II is a portable power drive unit that is normally bolted to a portable stand and used for threading pipe and bolt stock. This unit uses a mounted carriage with attached die-heads, reamer, and cutter, or coupled to a geared threader. Both power drives are intended for use in the shop or on the construction site due to their portability.

2. CLASSIFICATION. The power drives shall be classified by the following types as specified (see 7.3(b)):

Type I - Hand-held
Type II - Stand-mounted
Type III - Accessories only (without power drive)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3450

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3. SALIENT CHARACTERISTICS

3.1 Description.

3.1.1 Type I. The type I power drive shall consist of a portable frame, integral motor, control, and power supply cord. The power drive shall have gearing and provisions for mounting and supplying power to the die-head or other attachments.

3.1.2 Type II. The type II power drive shall consist of a housing, gearing, drive motor, control, and power supply cord. The power drive shall also have a spindle, rear centering chuck, front gripping chuck, support or extension arms, or way rods.

3.1.3 Type III. Unless otherwise specified (see 7.3(c)), type III shall consist of only the accessories or accessory sets for the type I or the type II power drive. The accessories shall comply with the requirements specified herein.

3.2 Interchangeability. All power drives of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.3 Design. Type I and II power drives shall be designed to be adapted by accessories to standard die-heads and geared threaders for cutting threads on pipe and bolt stock.

3.3.1 Type I. The type I power drive shall include a ring gear in which the die-head or adapter is mounted. This gear shall be mounted on sleeve or journal bearings and driven by the enclosed motor through a gear reduction. The gear shall weigh no more than 45 pounds (lbs.) (20.4 kilograms (kg)) with one die-head and adapter included.

3.3.2 Type II. The type II power drive shall be a self-contained fully enclosed assembly with necessary reduction gears and a cam-action rear centering device adjustable from 0.125 inch through 2 inches (3.2 millimeter (mm) through 50.8 mm). It shall have an adjustable positive grip drive-chuck capable of operation either with machine or geared die-heads. The drive-chuck shall be adjustable up to 2-inch (50.8-mm) diameter pipe size and shall be suitable for operation with a universal drive shaft. The drive-chuck shall be the wrenchless type. The drive-chuck jaws shall have easily replaceable jaw inserts for when the jaw teeth become worn. The type II power drive shall weigh no more than 100 lbs. (45 kg) (net weight without die-head, dies, adapters, stand, cutter, reamer, or other accessories).

3.4 Maintainability. Design and configuration of power drives, accessories, and adjustment features shall ensure that maintenance of operational capability and accuracy will be reduced to the simplest form or procedures. The operational maintenance will require a minimum of skills and common maintenance equipment. Clear, concise, and definitive instructions shall be provided for disassembly, repair, assembly, and adjustment. The power drive and component parts requiring lubrications, adjustment, cleaning, and operator maintenance shall be accessible without disassembly or removal of parts or components excluding such items as safety covers or guards.

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3.5 Performance.

3.5.1 Type I. The type I power drive shall turn the drop-head die-head with dies and cut quality threads on steel pipe 0.125 inch through 2 inches (3.2 mm through 50.8 mm) in diameter. The type I power drive shall turn the geared threader to produce threads on steel pipe 2.5 inch through 6 inches (63.5 mm through 152.4 mm) in diameter. With the proper dies, it shall cut metric threads on steel pipe or conduit 0.125 inch through 2 inches (3.2 mm through 50.8 mm) diameter. The type I power drive shall also turn dies to cut threads on steel bolt stock from 0.25 inch through 1 inch (6.3 mm through 25.4 mm) in diameter. When die-heads and dies are provided as part of the unit, the threads produced shall conform to American Society of Mechanical Engineers (ASME) B 18.2.1, "Square and Hex Bolts and Screws (Inch Series)", International Organization for Standardization (ISO) 7, "Pipe Threads Where Pressure-Tight Joints Are Made on the Threads", Part 1: "Dimensions, Tolerances and Designation", ISO 68, "General Purpose Screw Threads - Basic Profile", Part 2: "Inch Screw Threads" (Part 1: "Metric Screw Threads"), and ISO 261, "General Purpose Metric Screw Threads – General Plan", as applicable.

3.5.2 Type II. The type II power drive shall turn pipe and bolt stock for threading with die-head and dies mounted on a moveable carriage attached to the power drive. The type II power drive shall turn pipe for cutting and reaming with the cutter and reamer mounted on a moveable carriage. The type II power drive shall turn geared threaders short or long coupled to the power drive. Steel pipe from 0.125 inch through 2 inches (3.2 mm through 50.8 mm) in diameter and bolt stock from 0.25 inch through 2 inches (6.3 mm through 50.8 mm) in diameter shall be threaded by the carriage mount (machine mount dies). Steel pipe from 2.5 inch through 6 inches (63.5 mm through 152.4 mm) in diameter shall be threaded by geared threaders. When die-heads and dies are provided as part of the unit, the threads produced shall conform to ASME B 18.2.1, ISO 7-1, ISO 68, and ISO 261, as applicable.

3.6 Construction. The power drive units shall be constructed to withstand the strains and vibrations encountered in normal start-stop and turning or threading operations. The body shall enclose the power drives' moving parts and shall be of a suitable metal, either cast or fabricated.

3.6.1 Type I. The type I power drive body shall consist of a separate or integral housing for the drive motor mounted to the body.

3.6.1.1 Type II. The type II threader shall include a separate body or housing for the drive motor with an attached base with no less than four mounting holes for mounting to a stand. The type II housing shall include support or extension bars or way rods on which the attachments are to be mounted.

3.6.2 Motor. The motor shall be a reversible universal type rated at no less than 0.5 horsepower (373 watts) before gear reduction, 115/230 volts alternating current (VAC), 25 to 60 Hertz. The type II spindle speed shall be not less than 35 revolutions per minute standard. Unless otherwise specified (see 7.3(d)), the internal motor wiring shall be connected for 115 VAC operations.

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3.6.2.1 Power supply cord. The power supply cord shall be not less than 8 feet (2.4 meter (m)) long, type stranded, oil-resistant compound (SO) or stranded, thermoplastic (ST), number 16 American Wire Gage (AWG) or larger, copper, stranded, with 3 conductors. The plug shall be a 3-wire grounding type. The power cord shall otherwise conform to National Fire Protection Association (NFPA) 70, "National Electrical Code, Article 400 -- Flexible Cords and Cables". Grounding shall conform to NFPA 70, "National Electrical Code, Article 250 -- Grounding".

3.6.2.2 Motor control. The motor control shall be a 3-position switch rated for the motor full load current. The switch for type I shall be the momentary type. The switch may be a combination directional control and on-off momentary type. The switch shall be mounted convenient to the operator and protected against accidental starting. The switch shall be marked to indicate FORWARD, OFF, and REVERSE positions. Type II power drive shall include a momentary OFF and ON foot switch (with shield) in addition to the forward-off-reverse switch. The switch shield shall be oversized to accommodate safety-toe shoes. A guard to prevent unintentional operation of the treadle shall be included.

3.7 Power drive attachments, components, and material. Type I and II power drives, when augmented from options covered in 3.8 through 3.10.7, can be tailored to perform a variety of functions. Augmented items shall be considered part of the power drive unit when selected for a specific contract or order. Augmented items shall be compatible with the complete unit for fit and function.

3.8 Threading attachments for type I power drives. When specified (see 7.3(e)), threading attachments for type I power drives shall include the die-heads, dies, die-head adapter (for bolt threading die-head), adapter (gear threader), support (torque) arm, and carrying cases. Dies for pipe threading shall be the die segment insert type. Bolt dies for the drop-head may be the button type.

3.8.1 Die-head and dies. Die-head and dies include the drop-head, die-head with alignment guide, and die inserts for threading pipe and bolt stock in the inch and ISO metric system (pipe threads 0.125 inch through 2 inches (3.2 mm through 50.8 mm)) and 0.630 inch through 1.900 inches (16 mm through 48.3 mm) in diameter). Threads produced shall conform to ASME B 18.2.1 or ISO 7-1, as appropriate.

3.8.2 Adapters. Adapters shall adjust the bolt threading die-head and the gear threader to the type I power drive.

3.8.3 Support arm. The support arm shall clamp onto the work to resist the torque of the type I power drive when threading.

3.8.4 Carrying case. Carrying cases shall be provided for the type I power drive, adapters, and support arm, and for pipe and bolt die-heads and dies. The cases shall include latches and partitions to separate the contents and keep them in location.

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3.9 Threading attachments for type II power drives. When specified (see 7.3(f)), threading attachments for type II power drives shall include a carriage with lever arm, reamer, cutter, dies, and die-heads (machine die-heads).

3.9.1 Carriage. The carriage mounts on the support arm(s) of the power drive and provides a moveable base for the reamer, cutter, and die-head. Increased mechanical leverage is provided by the lever arm to move the carriage laterally for threading and reaming operations.

3.9.2 Reamer. The reamer is attached to the carriage at a pivot point to permit rotating the reamer into position for reaming the pipe. The reamer shall be a straight-fluted cone type.

3.9.3 Cutter. The cutter is attached to the carriage at a pivot point to permit rotating the cutter into position for cutting the pipe. The cutter shall be a full floating wheel type.

3.9.4 Dies and die-head. Dies and die-heads include the self-opening and quick-opening (universal) die-heads with die segment inserts. Automatic opening die-heads must be manually opened in size 0.375 inch (9.5 mm) iron pipe size (IPS) and smaller.

3.9.4.1 Self-opening die-head. The closing movements shall be controlled manually; opening shall be controlled both manually and automatically when the full thread is complete. Provisions for adjustment of dies to produce oversize and undersized threads shall be included.

3.9.4.2 Quick-opening (universal) die-head. The opening and closing movements shall be controlled manually and the die-head shall include provisions for adjustment of dies to produce oversize and undersize threads.

3.10 Miscellaneous threading components, attachments, and accessories. Items selected from this group are intended as options to augment either the type I or type II power drive or selected individually for replacement of broken or worn out items of a unit or to expand the function of an existing unit. When individual items are requisitioned and are not part of an order with a power drive, the make and model of the unit the item is intended to fit must be indicated to ensure compatibility (see 7.3(g)). When specified (see 7.3(h)), items selected include geared threaders with dies, drive bar, nipplechuck and adapters for threading short and close pipe nipples and bolts, a tool stand with tray, pipe support stand, oiler, cutting oil, and ratchet with handle.

3.10.1 Geared pipe threader. The geared pipe threader is powered by a separate power drive through a short or long drive shaft. The threader is adjustable from a diameter of 2.5 inches through 4 inches (63.5 mm through 102 mm) or 4 inches through 6 inches (102 mm through 152.4 mm) with one set of dies in each range. The geared threader is attached to the drive unit for short couple or supported on the pipe to be threaded for long couple. A safety device to prevent accidental jamming and an integral adjustable guide to maintain pipe alignment shall be included. The geared threaders shall be adjustable for oversize and undersize National Pipe Straight Mechanical (NPSM) threads.

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3.10.2 Dies. The dies include pipe and bolt dies in the inch and International System of Units (SI) metric systems to fit the self-opening and quick-opening (universal) die-heads and geared threader. Dies shall be of high speed steel and shall cut threads conforming to ASME B 18.2.1, ISO 7-1, or ISO 261, as applicable. Dies shall be hardened to a hardness range of Rockwell C58 to C66.

3.10.3 Nipplechuck and adapters for nipples and bolts. The nipplechuck is attached to the main chuck of the power drive. Nipple and bolt adapters are attached to nipplechuck for holding and turning short and close nipples and bolt stock for threading. Threads are cut by the dies provided for the die-heads. Threads cut by adapters shall conform to ASME B 18.2.1, ISO 7-1, or ISO 261, as applicable.

3.10.4 Tool stand. The tool stand shall be designed to support the type II power drive during normal threading operations without buckling, distorting, or moving. Provisions for bolting the power drive to the stand shall be included. The height of the stand shall permit threading operations by an operator who works in the standing position. A tray or shelf shall be mounted between the legs to afford additional rigidity to the stand and permit storage of the tools and parts. The stand shall be designed for easy assembly and disassembly.

3.10.5 Pipe support stand. The adjustable stand shall be capable of supporting pipe without tipping or rocking, yet allow the pipe to rotate over the rollers mounting in the crotch of the stand. The stand shall have a vertical adjustment range from 24 to 32 inches (610 mm to 813 mm) and be a suitable height for operating with type II power drives while mounted on the tool stand and close coupled to the gear threader. Capacity of the stand shall be for 10-inch (254-mm) diameter pipe.

3.10.6 Oiler. When threading, the oiler for oiling the dies shall include a hand pump attached to a pan by an oil resistant hose (not less than 4 feet (1.2 m) long). The chip pan serves as an oil reservoir and catch for oil run-off to prevent oil waste and permit the reuse of the oil. The pan shall provide screening of oil run-off before the oil is drawn from the pan by the pump. The pan shall include a hinged cover.

3.10.7 Ratchet and handle. The ratchet shall have a 1-inch (25.4-mm) square drive socket to fit a pinion shaft 0.96875 inch (24.6 mm) square on the geared pipe threader. The square socket drive hole shall extend through the ratchet drum so that the ratchet may be placed on the pinion shaft through either side of the drum. The ratchet shall provide forward and reverse operation either by control of the ratchet pawls or by inserting the driven shaft in the socket-drive-hole on the opposite side of the ratchet drum. Ratchet and handle shall withstand the force on the end of the handle to produce 200 foot-pounds (ft.-lbs.) (271 Newton meters (Nm)) of torque on a shaft inserted in the square drive socket. The handle and ratchet combined shall measure between 30 and 36 inches (762 mm and 914.4 mm) long with the handle not less than 27 inches (686 mm) long. The ratchet drum, frame, and pawl assembly shall not fail when subjected to not less than 400 ft.-lbs. (542 Nm) of torque. Unless otherwise specified (see 7.3(i)), the ratchet and handle shall weigh not less than 9 lbs. (4 kg). The handle shall be threaded on one end for attaching to the ratchet and to permit removal of the handle from the ratchet.

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3.10.8 Cutting oil. The thread cutting oil shall provide cooling to tools and work and aid in the cutting of smooth threads by reducing friction. The oil shall be nonfoaming, nontoxic, and noncorrosive to steel, copper, and tool parts.

3.11 Instruction plates. The power drives shall be equipped with instruction plates that are suitably located and describe any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material that will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 0.125 inch (3.2 mm) in diameter.

3.12 Identification plate. Unless otherwise specified (see 7.3(j)), the contracting officer will furnish the government inspector with the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix the plates in a conspicuous place on each unit, assembly, or subassembly, and parts as directed by the government inspector. Screws, rivets, or bolts shall be used to affix the plates. Nomenclature shall be "DRIVE-POWER, PORTABLE: FOR PIPE AND BOLT THREADING, HAND HELD, ELECTRIC", or "DRIVE-POWER, PORTABLE: FOR THREADING, STAND MOUNTED, ELECTRIC", as appropriate.

3.13 Painting. Unless otherwise specified (see 7.3(k)), painting and finishing of the power drive shall be in accordance with the contractor's standard practice provided the following minimum criteria are met or exceeded. All surfaces to be painted shall have been cleaned to be free of all foreign matter prior to painting.

3.14 Spare parts and maintenance tools. When specified (see 7.3(l)), spare parts and maintenance tools shall be furnished.

3.15 Gears. Spur, pinion, worm, and bevel gears shall be machined in the inch or SI system (manufacturer's option) and shall otherwise conform to ISO 53, "Cylindrical Gears for General and Heavy Engineering - Standard Basic Rack Tooth Profile", American Gear Manufacturers Association (AGMA) 1003, "Tooth Proportions for Fine-Pitch Spur and Helical Gearing", AGMA 2005, "Design Manual for Bevel Gears", AGMA 6010, "Standard for Spur, Helical, Herringbone and Bevel Enclosed Drives", AGMA 6030, "Design of Industrial Double-Enveloping Wormgears", or AGMA 6034, "Practice for Enclosed Cylindrical Wormgear Speed Reducers and Gearmotors", as appropriate.

3.16 Workmanship. Workmanship of the power drive and accessories shall be of the best quality prevailing among manufacturers of this type of equipment. The items shall be free from all imperfections that will adversely affect the general appearance, function, and serviceability.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of Institute of Electrical and Electronic Engineers (IEEE) SI 10, "American National Standard for Use of the International System of Units (SI): The Modern Metric System", and all other requirements of this CID, including form, fit, and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

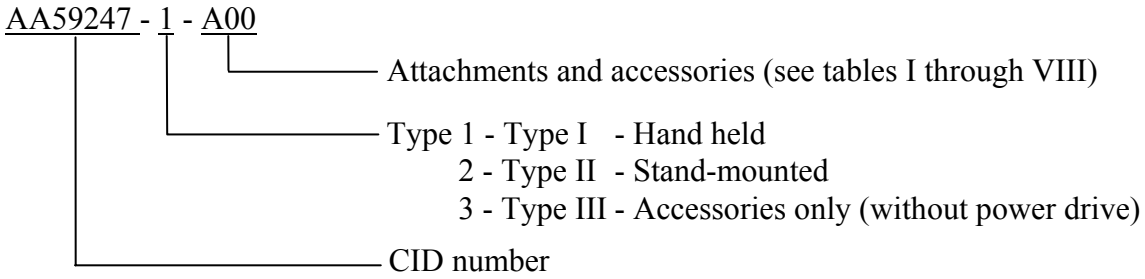
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(m)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 AGMA. Copies of AGMA standards may be obtained from the American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, VA 22314. Electronic copies of AGMA documents may be obtained from http://www.agma.org/.

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7.2.2 ASME standards. Copies of ASME standards may be obtained from the ASME International, Three Park Avenue, New York, NY 10016-5990. Electronic copies of ASME standards may be obtained from http://www.asme.org/.

7.2.3 FAR. Copies of the Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.4 IEEE standards. Copies of IEEE standards may be obtained from the Institute of Electrical and Electronics Engineers, 3 Park Avenue, 17th Floor, New York, NY 10016-5997. Electronic copies of IEEE standards may be obtained from http://www.ieee.org/.

7.2.5 ISO standards. Copies of the ISO standards may be obtained from American National Standards Institute (ANSI), 25 West 43rd Street, 4th Floor, New York, NY 10036. Electronic copies of ISO standards may be obtained from http://web.ansi.org/.

7.2.6 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101. Electronic copies of NFPA standards may be obtained from http://www.nfpa.org/.

7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Type of power drive (see 2).
c. When type III is other than specified (see 3.1.3).
d. When motor wiring is other than specified (see 3.6.2).
e. Threading attachments, if required, for type I (see 3.8).

    1. Attachment set for type I or II (see table I).
    2. Pipe die-head (see 3.8.1 and tables I through IV).
    3. Pipe dies (see 3.8.1 and tables II through VIII).
    4. Bolt die-head (see 3.8.2 and tables V and VIII).
    5. Bolt die-head adapter (see 3.8.2 and table V).
    6. Geared threader adapter (see 3.8.2 and table VIII).
    7. Support (torque) arm (see 3.8.3 and table VI).
    8. Carrying cases (see 3.8.4 and table VI).

f. Threading attachments, if required, for type II (see 3.9).

    1. Carriage (see 3.9.1 and tables VII and VIII).
    2. Reamer (see 3.9.2 and table VII).
    3. Cutter (see 3.9.3 and table VII).
    4. Dies (see 3.9.4 and table VII).
    5. Die-head (see 3.9.4.1and 3.9.4.2, and table VII).

g. Make and model of the unit receiving replacement parts (see 3.10).

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h. Miscellaneous threading components, attachments, and accessories, if required (see 3.10).

1. Geared pipe threader (see 3.10.1 and table VIII).
2. Drive bar for geared threader (see 3.10.1 and table VIII).
3. Dies, inch or metric (see 3.10.2 and table VIII).
4. Nipplechuck and adapters for nipples and bolts (see 3.10.3 and table VIII).
5. Tool stand (see 3.10.4 and table VIII).
6. Pipe support stand (see 3.10.5 and table VIII).
7. Oiler (see 3.10.6 and table VIII).
8. Ratchet and handle (see 3.10.7 and table VIII).
9. Cutting oil (see 3.10.8 and table VIII).

i. When ratchet and handle weigh less than 9 lbs. (4 kg) (see 3.10.7).
j. When identification plates are furnished to the government other than by the contracting officer (see 3.12).
k. When painting and finishing of the power drive is other than contractor's standard practice (see 3.13).
l. When spare parts and maintenance tools are furnished (see 3.14).
m. Preservation, packing, and marking (see 6.1).

7.4 Subject term (key word) listing.

cutting
die-head
dies
geared threader
reaming

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TABLE I. Combination of attachment sets for type I and type II power drives.

PIN code	Item description
A00	Power drive with no attachments
Attachment sets combination - type I power drive:	
A01	Attachment sets B01 and C01
A02	Attachment sets B01 and D01
A03	Attachment sets B01 and E01
A04	Attachment sets B01 and F01
A05	Attachment sets C01 and D01
A06	Attachment sets C01 and E01
A07	Attachment sets C01 and F01
A08	Attachment sets D01 and E01
A09	Attachment sets D01 and F01
A10	Attachment sets E01 and F01
A11	Attachment sets B01, C01, and D01
A12	Attachment sets B01, C01, and E01
A13	Attachment sets B01, C01, and F01
A14	Attachment sets B01, D01, and E01
A15	Attachment sets B01, D01, and F01
A16	Attachment sets B01, E01, and F01
A17	Attachment sets C01, D01, and E01
A18	Attachment sets C01, D01, and F01
A19	Attachment sets C01, E01, and F01
A20	Attachment sets D01, E01, and F01
A21	Attachment sets B01, C01, D01, and E01
A22	Attachment sets B01, C01, D01, and F01
A23	Attachment sets B01, C01, E01, and F01
A24	Attachment sets B01, D01, E01, and F01
A25	Attachment sets C01, D01, E01, and F01
A26	Attachment sets B01, C01, D01, and F01
Attachment set combination - type II power drive:	
A027	Attachment sets G01 and H01

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TABLE II. Pipe (inch) threading attachments for type I power drive.

PIN code	Item name	Type[1]	Size[2]
B01	Complete set of all accessories in the table		
B02	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	0.125-inch - 27 TPI
B03	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	0.25-inch - 18 TPI
B04	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	0.375-inch - 18 TPI
B05	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	0.5-inch - 14 TPI
B06	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	0.75-inch - 14 TPI
B07	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	1.0-inch - 11.5 TPI
B08	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	1.25-inch - 11.5 TPI
B09	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	1.5-inch - 11.5 TPI
B10	Die-head, pipe, high speed, complete	Right hand NPT (segment insert)	2.0-inch - 11.5 TPI

[1]NPT = national pipe thread.
[2]TPI = threads per inch.

TABLE III. Conduit (metric) threading attachments for type I power drive.

PIN code	Item name	Type[1]	Diameter-thread pitch[2]
C01	Complete set of all accessories in the table		
C02	Die-head, conduit, complete	Conduit, IS, taper R-1.5 (segment insert)	16 mm - 1.5 mm TP
C03	Die-head, conduit, complete	Conduit, IS, taper R-1.5 (segment insert)	20 mm - 1.5 mm TP
C04	Die-head, conduit, complete	Conduit, IS, taper R-1.5 (segment insert)	25 mm - 1.5 mm TP
C05	Die-head, conduit, complete	Conduit, IS, taper R-1.5 (segment insert)	32 mm - 1.5 mm TP
C06	Die-head, conduit, complete	Conduit, IS, taper R-1.5 (segment insert)	40 mm - 1.5 mm TP

[1]IS = international standards; R = taper, external.
[2]TP = thread pitch.

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TABLE IV. Pipe (metric) threading attachments for type I power drive.

PIN code	Item name	Die thread type, and designator	Pipe size nominal (mm)	Thread designator (inch)	Thread pitch (mm)
D01	Complete set of accessories in table				
D02	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-0.125	6	0.125	0.907
D03	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-0.25		0.25	1.337
D04	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-0.375	10	0.375	1.337
D05	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-0.5	15	0.5	1.814
D06	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-0.75	20	0.75	1.814
D07	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-1.0	25	1.0	2.309
D08	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-1.25	32	1.25	2.309
D09	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-0.5	40	1.5	2.309
D10	Die-head, pipe, high speed, complete	Pipe (segment insert) IS, taper, R-2.0	50	2.0	2.309

TABLE V. Bolt (inch) threading attachments for type I power drive.

PIN code	Item name	Type[1]	Size
E01	Complete set of all accessories in the table		
E02	Die-head, bolt, complete	RH, UNC (button or insert)	0.25-inch - 20 TPI
E03	Die-head, bolt, complete	RH, UNC (button or insert)	0.3125-inch - 18 TPI
E04	Die-head, bolt, complete	RH, UNC (button or insert)	0.375-inch - 16 TPI
E05	Die-head, bolt, complete	RH, UNC (button or insert)	0.4375-inch - 14 TPI
E06	Die-head, bolt, complete	RH, UNC (button or insert)	0.5-inch - 13 TPI
E07	Die-head, bolt, complete	RH, UNC (button or insert)	0.5625-inch - 12 TPI
E08	Die-head, bolt, complete	RH, UNC (button or insert)	0.625-inch - 11 TPI
E09	Die-head, bolt, complete	RH, UNC (button or insert)	0.75-inch - 10 TPI
E010	Die-head, bolt, complete	RH, UNC (button or insert)	0.875-inch - 9 TPI
E011	Die-head, bolt, complete	RH, UNC (button or insert)	1.0-inch - 8 TPI
E012	Die-head adapter	Bolt die-head	

[1]RH = right hand, UNC = unified national course.

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TABLE VI. Miscellaneous attachments for type I power drive.

PIN code	Item name	Type	Size
F01	Complete set of all accessories in the table		
F02	Support arm	Clamp-on	2-inch capacity
F03	Adapter	Gear threader	Standard
F04	Carrying case	Metal	For type I power drive and attachments
F05	Carrying case	Metal	For type I dies and die-heads

TABLE VII. Threading attachments for type II power drive.

PIN code	Item name	Type	Size[1]
G01	Complete set of all accessories in the table		
G02	Carriage with lever arm	For die-head, reamer, and cutter	Standard
G03	Reamer	Straight flute, carriage mount	See 3.9.2
G04	Cutter	Wheel, floating, carriage mount	See 3.9.3
G05	Die-head (inch)	Self-opening, RH	0.125- to 2.0-inch IPS
G06	Die-head (inch)	Quick - opening (universal) RH	0.125- to 2.0-inch IPS
G07	Die, high speed (set)	Pipe, RH, NPT	0.25- and 0.375-inch - 18 TPI
G08	Die, high speed (set)	Pipe, RH, NPT	0.5 -and 0.75-inch - 14 TPI
G09	Die, high speed (set)	Pipe, RH, NPT	1.0 and 2.0-inch - 11.5 TPI
G010	Die-head (metric)	Self-opening, RH	6 mm - 50 mm IPS
G011	Die-head (metric)	Quick - opening (universal) RH	6 mm - 50 mm IPS
G012	Die, high speed (set)	Pipe, IS, taper, R-0.25 and R-0.375	8 mm - 10 mm
G013	Die, high speed (set)	Pipe, IS, taper, R-0.5 and R-0.75	15 mm - 20 mm
G014	Die, high speed (set)	Pipe, IS, taper, R-1 to R-2	25 mm - 50 mm
G015	Die-head (metric)	Bolts, IS	6 mm through 26 mm
G016	Die-head (metric)	Bolts, IS	27 mm through 52 mm

[1]IPS = iron pipe size.

NOTES:

1. The above dies shall fit both the self-opening, and quick-opening (universal) die-heads.
2. Three sets of dies are required to cover the range from 0.25-inch through 2-inch diameters. Sets of dies are adjustable through diameter ranges 0.25-inch through 0.375-inch, 0.5-inch through 0.75-inch, and 1.0-inch through 2-inches or ranges 8-10 mm, 15-20 mm, 25-50 mm diameter with either of the above die-heads.
3. Dies in one size range shall be provided in matched sets.

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TABLE VIII. Miscellaneous threading components, attachments, and accessories.

PIN code	Item name	Type[1]	Size range (nominal)
H01	Complete set of all accessories in the table		
H02	Pipe threader	Geared, adjustable	2.5- to 4.0-inch diameter
H03	Pipe threader	Geared, adjustable	4.0- to 6.0-inch diameter
H04	Pipe threader (metric)	Geared, adjustable	65 mm through 100 mm
H05	Pipe threader (metric)	Geared, adjustable	125 mm through 150 mm
H06	Die, geared pipe threader (set)	High speed, RH, NPSM, NPT	2.5 through 4.0 inch - 8 TPI
H07	Die, geared pipe threader (set)	High speed, RH, NPSM, NPT	4.0 through 6.0 inch - 8 TPI
H08	Die, geared pipe threader	High speed, IS, taper	65 mm through 100 mm dia.
H09	Die, geared pipe threader (metric)	High speed, IS, taper	100 mm through 150 mm dia.
H010	Drive bar, geared threader	Close couple	Standard, 1.0-inch square
H011	Die, bolt (inch) (set)	RH, UNC, closed	0.375-inch - 16 TPI
H012	Die, bolt (inch) (set)	RH, UNC, closed	0.4375-inch - 14 TPI
H013	Die, bolt (inch) (set)	RH, UNC, closed	0.5-inch - 13 TPI
H014	Die, bolt (inch) (set)	RH, UNC, closed	0.5625-inch - 12 TPI
H015	Die, bolt (inch) (set)	RH, UNC, closed	0.625-inch - 11 TPI
H016	Die, bolt (inch) (set)	RH, UNC, closed	0.75-inch - 10 TPI
H017	Die, bolt (inch) (set)	RH, UNC, closed	0.875-inch - 9 TPI
H018	Die, bolt (inch) (set)	RH, UNC, closed	1.0-inch - 8 TPI
H019	Die, bolt (inch) (set)	RH, UNC, closed	1.25-inch - 7 TPI
H020	Die, bolt (inch) (set)	RH, UNC, closed	1.5-inch - 6 TPI
H021	Die, bolt (inch) (set)	RH, UNC, closed	1.75-inch - 5 TPI
H022	Die, bolt (inch) (set)	RH, UNC, closed	2.0-inch - 4.5 TPI
H023	Die, bolt (metric) (set)	RH, IS, closed	6 mm - 1 mm TP
H024	Die, bolt (metric) (set)	RH, IS, closed	8 mm - 1.25 mm TP
H025	Die, bolt (metric) (set)	RH, IS, closed	10 mm - 1.5 mm TP
H026	Die, bolt (metric) (set)	RH, IS, closed	12 mm - 1.75 mm TP
H027	Die, bolt (metric) (set)	RH, IS, closed	16 mm - 2 mm TP
H028	Die, bolt (metric) (set)	RH, IS, closed	18 mm - 2.5 mm TP
H029	Die, bolt (metric) (set)	RH, IS, closed	20 mm - 2.5 mm TP
H030	Die, bolt (metric) (set)	RH, IS, closed	22 mm - 2.5 mm TP

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TABLE VIII. Miscellaneous threading components, attachments, and accessories - Continued.

PIN code	Item name	Type[1]	Size range (nominal)
H031	Die, bolt (metric) (set)	RH, IS, closed	24 mm - 3 mm TP
H032	Die, bolt (metric) (set)	RH, IS, closed	30 mm - 3.5 mm TP
H033	Die, bolt (metric) (set)	RH, IS, closed	36 mm - 4 mm TP
H034	Die, bolt (metric) (set)	RH, IS, closed	42 mm - 4.5 mm TP
H035	Die, bolt (metric) (set)	RH, IS, closed	48 mm - 5 mm TP
H036	Die, bolt (metric) (set)	RH, IS, closed	50 mm - 5 mm TP
H037	Chuck, nipple and bolt	Short and close nipple & bolt	0.5 to 2.0 inch
H038	Adapter, nipple	RH, NPT	0.375-inch - 18 TPI
H039	Adapter, bolt	RH, UNC	0.375-inch - 16 TPI
H040	Adapter, bolt	RH, UNC	0.4375-inch - 14 TPI
H041	Adapter, bolt	RH, UNC	0.5-inch - 13 TPI
H042	Adapter, bolt	RH, UNC	0.5625-inch - 12 TPI
H043	Adapter, bolt	RH, UNC	0.625-inch - 11 TPI
H044	Adapter, bolt	RH, UNC	0.75-inch - 10 TPI
H045	Adapter, bolt	RH, UNC	0.875-inch - 9 TPI
H046	Adapter, bolt	RH, UNC	1.0-inch - 8 TPI
H047	Adapter, bolt	RH, UNC	1.25-inch - 7 TPI
H048	Adapter, bolt	RH, UNC	1.5-inch - 6 TPI
H049	Adapter, bolt	RH, UNC	1.75-inch - 5 TPI
H050	Adapter, bolt	RH, UNC	2.0-inch - 4.5 TPI
H051	Adapter, bolt (metric)	RH, IS	6 mm - 1 mm TP
H052	Adapter, bolt (metric)	RH, IS	8.25 mm - 1 mm TP
H053	Adapter, bolt (metric)	RH, IS	10 mm - 1.5 mm TP
H054	Adapter, bolt (metric)	RH, IS	12 mm - 1.75 mm TP
H055	Adapter, bolt (metric)	RH, IS	16 mm - 2 mm TP
H056	Adapter, bolt (metric)	RH, IS	18 mm - 2.5 mm TP
H057	Adapter, bolt (metric)	RH, IS	20 mm - 2.5 mm TP
H058	Adapter, bolt (metric)	RH, IS	22 mm - 2.5 mm TP
H059	Adapter, bolt (metric)	RH, IS	24 mm - 3 mm TP
H060	Adapter, bolt (metric)	RH, IS	30 mm - 3.5 mm TP
H061	Adapter, bolt (metric)	RH, IS	36 mm - 4 mm TP

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TABLE VIII. Miscellaneous threading components, attachments, and accessories - Continued.

PIN code	Item name	Type[1]	Size range (nominal)
H062	Adapter, bolt (metric)	RH, IS	42 mm - 4.5 mm TP
H063	Adapter, bolt (metric)	RH, IS	48 mm - 5 mm TP
H064	Adapter, bolt (metric)	RH, IS	50 mm - 5 mm TP
H065	Tool stand	Adjustable, power-drive	Standard
H066	Support stand	Pipe	6.0-inch pipe diameter
H067	Oiler	Hand operated, with pan	1 gallon capacity min.
H068	Ratchet and handle	Reversible	1.0-inch drive
H069	Oil	Thread cutting	5 gallons

[1]NPSM = national pipe straight mechanical.
NOTE: Dies for threading nipples or bolts are to fit the self-opening and quick-opening die-heads of table VII.

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - MC
Air Force - 99

Review Activities:
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS6

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